

02464 Artificial Intelligence and Human Cognition

Overview

Teachers



Ivana

Tobias



Teaching assistants



Mario



Hanlu

Teaching assistants



Kathrine



Aliaksandr

Motivation

- AI systems inspire models of human cognition
 - Perception and action as Bayesian inference
- How can we compare human and computers?
 - Human intelligence vs artificial intelligence
 - Learning
 - Memory
- Can we think of AI as a reverse engineering problem?
 - Not generally
 - Neural networks, language models and automatic speech recognition
- AI systems learn from humans
 - Mimicking human behavior
 - Image classification (typically human labels, not ground truth)
- AI systems are evaluated by humans
 - Language - who is the judge of how well GPT works?

General info

- Go to 'overview' under content on DTU Learn for general course info

Exam

- The final written exam is on 16 December 2025. We do not know the time or place for the exam yet. We will announce it when we know.
- Your grade in the course will be based 80% on the final exam, and 20% on the submitted exercises (10% per exercise).

Reports / exercises

- The exercises are a key part of the course curriculum and will be part of the exam. Completing the exercises and lectures will prepare you for the exam questions.
- There are two mandatory exercises in the course
 - (1 for Tobias' half, 1 for Ivana's)
- During some of the exercises you will solve coding problems. You will need to have Matlab installed for the exercise in week 3. Coding in Matlab will not be part of the exam.

Course structure

See 'overview' under 'content' on DTU Learn

Topics	Teacher	Teaching week
Introduction to cognitive science	Tobias	2 & 7
Perception as Bayesian Inference	Tobias	3-4
Neural coding of visual information	Tobias	5-6
Motor control	Ivana	8
Social cognition	Ivana	9-10
Embodied cognition and psychophysiology	Ivana	1, 11-12

Intro to Cognitive Science - week 2 & 7

How can we compare humans and computers?

- Intelligence
 - How do we define intelligence?
- Computation
 - Marr's three levels
- Learning
 - Machine learning paradigms as models of human learning
- Memory
 - How do we measure human memory capacity?

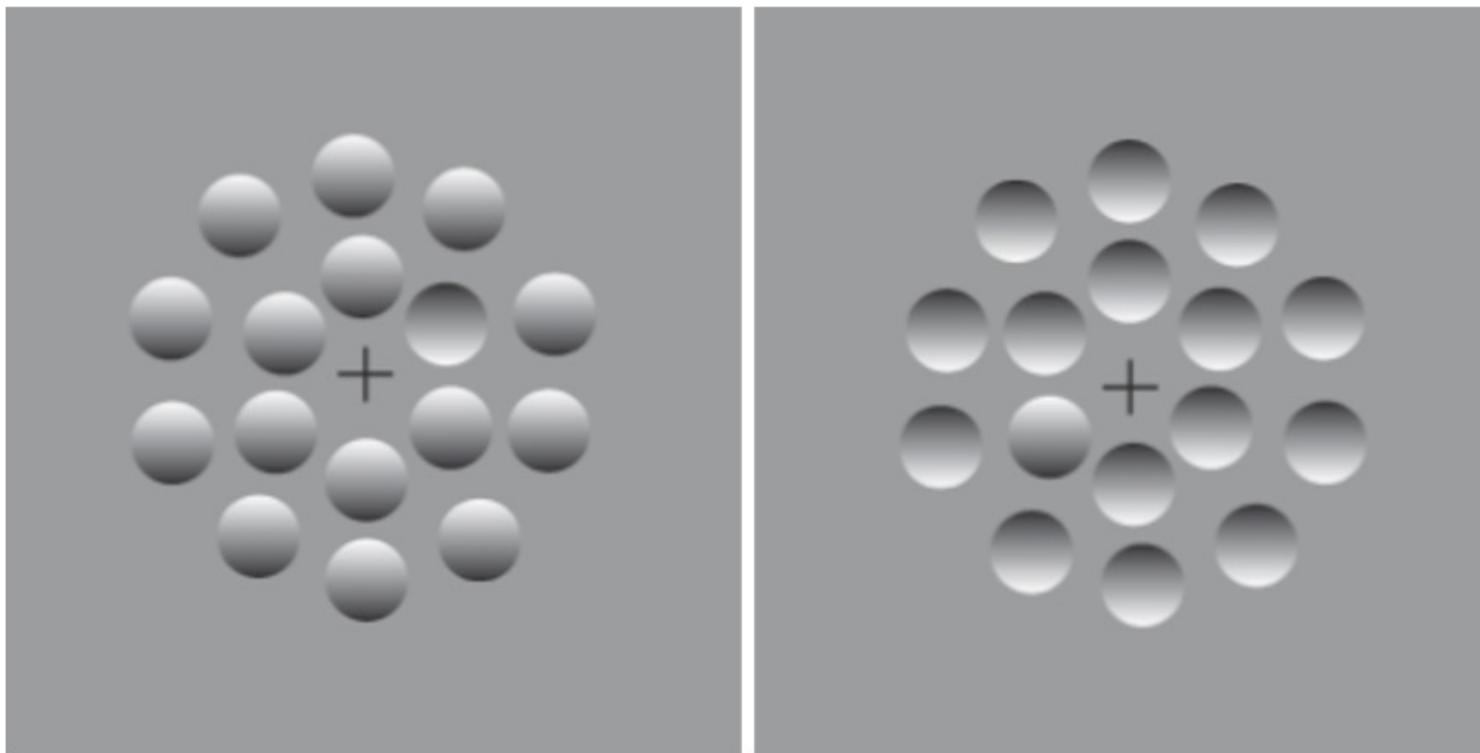
Perception/Bayes – week 3-4

What's the fuzz?!

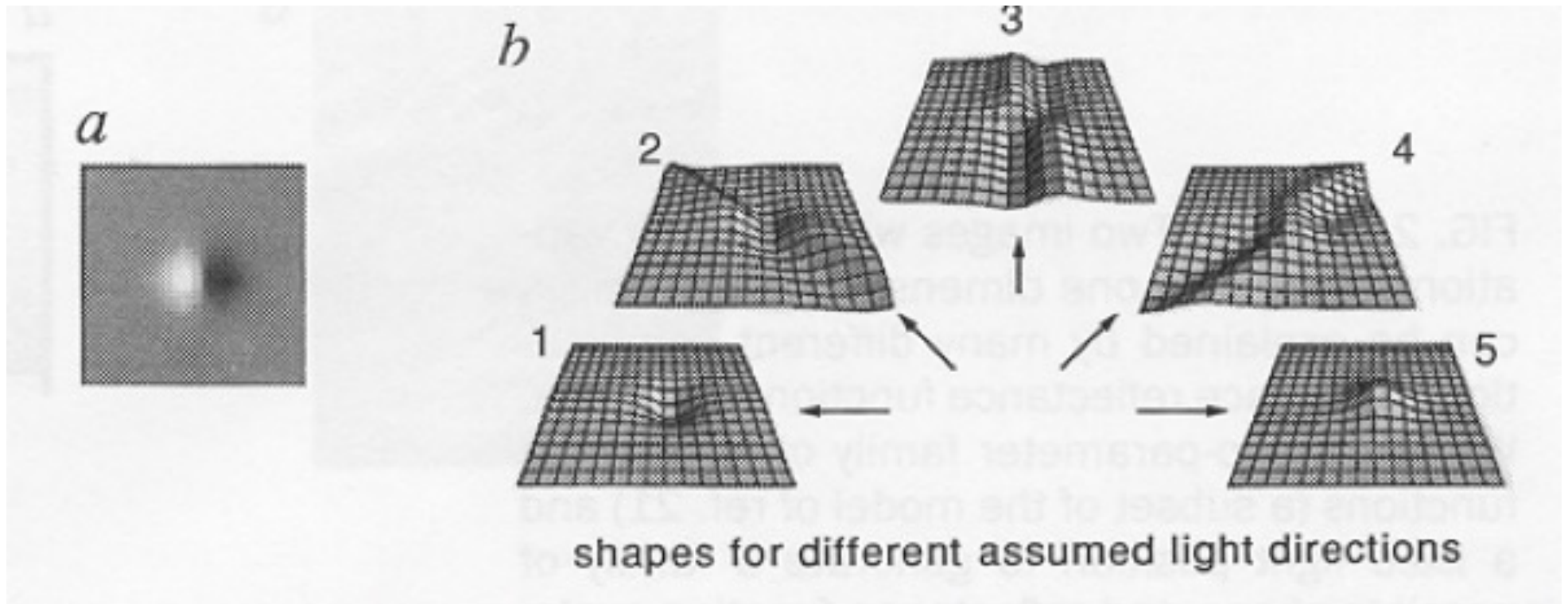
$$P(w \mid d) = \frac{P(d \mid w)P(w)}{P(d)}$$

- The problem of *any* observer
- The problem of perception
 - Probability of the world given sensory *data*
- Likelihood contains a model of the world
 - Not enough information
 - Prior adds information

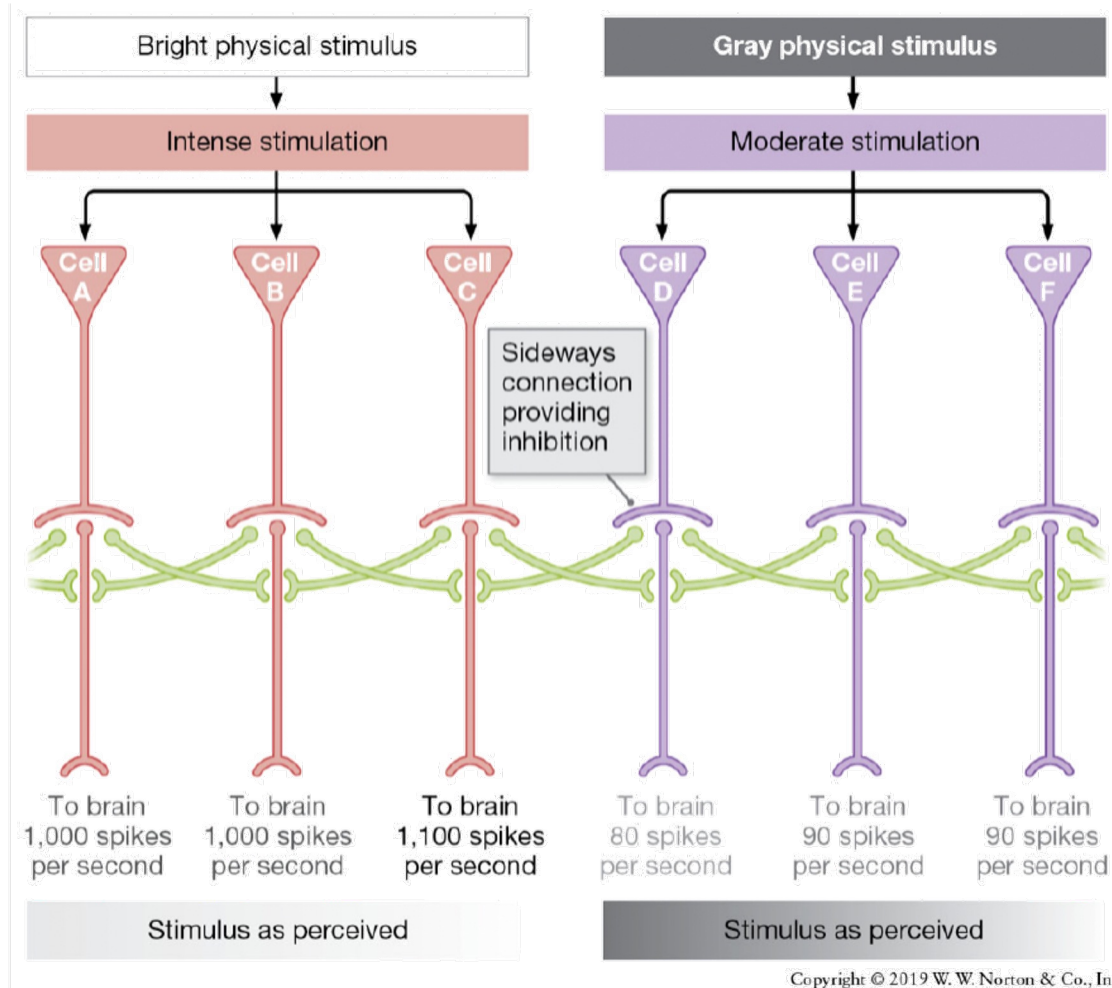
Perception/Bayes – week 3-4



Perception/Bayes – week 3-4



Neural coding – week 5-6



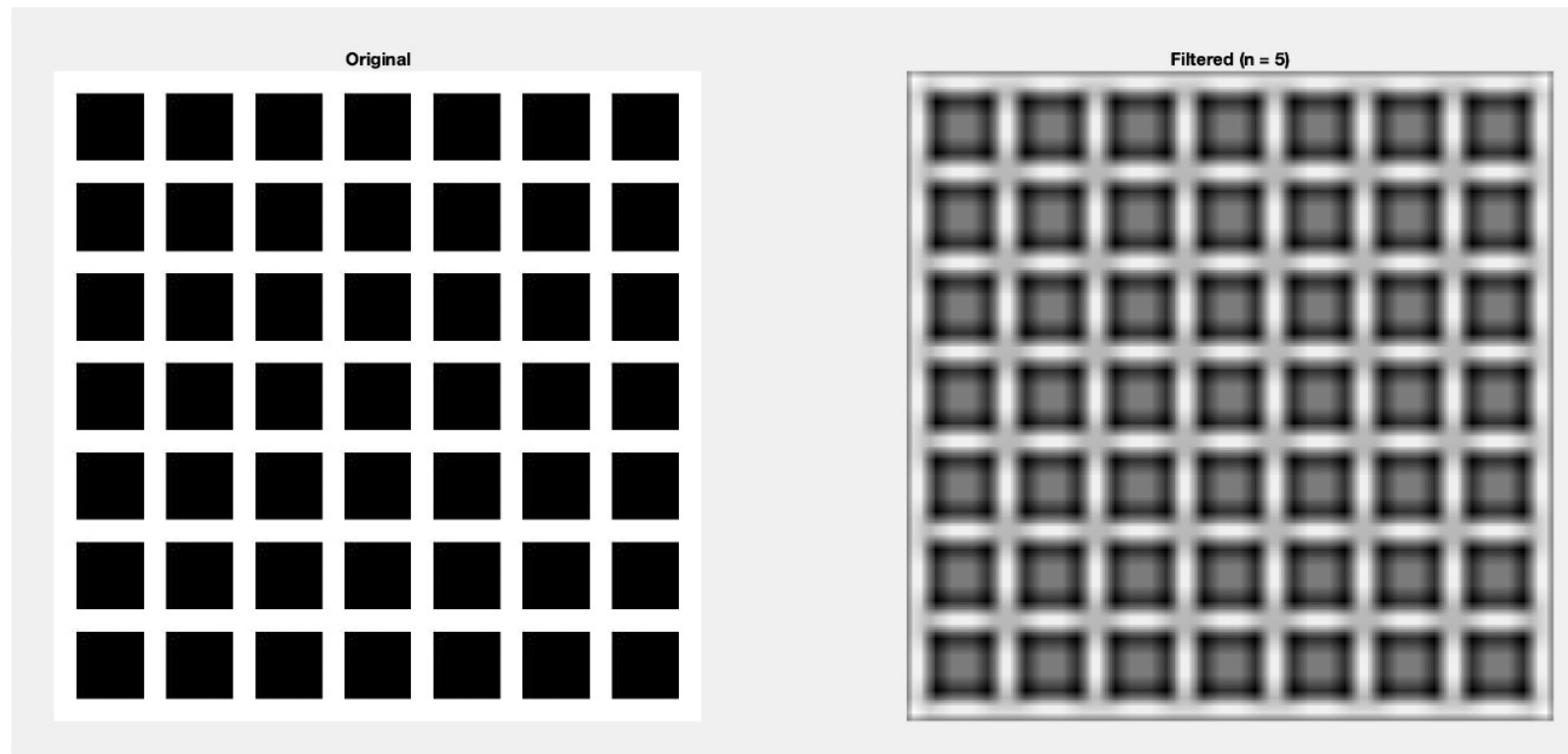
Neural coding – week 5-6

Mona Lisa



Neural coding – week 5-6

The Hermann Grid



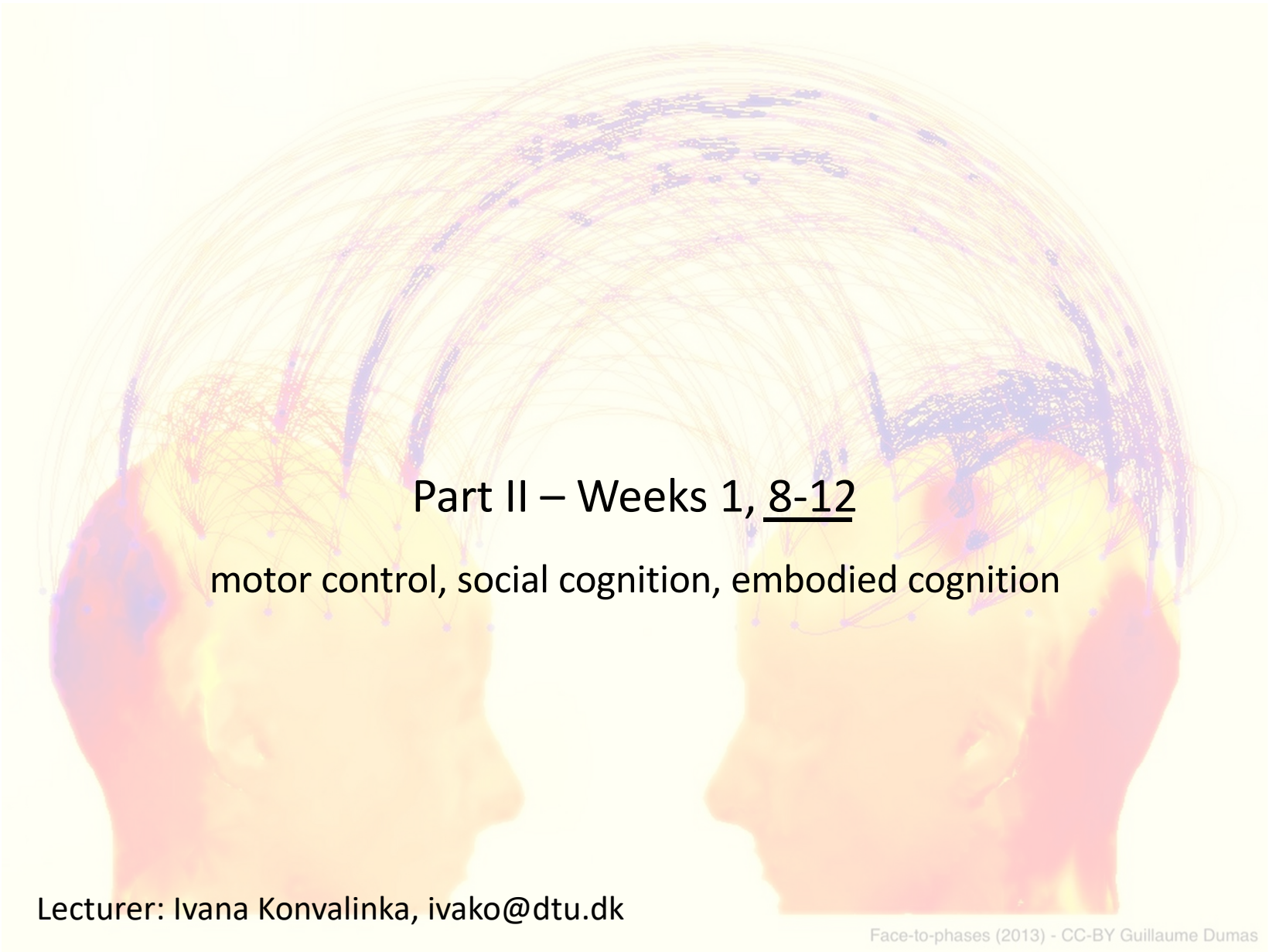
Neural coding – week 5-6

Image PCA

Who's who?

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Part II – Weeks 1, 8-12

motor control, social cognition, embodied cognition

Lecturer: Ivana Konvalinka, ivako@dtu.dk

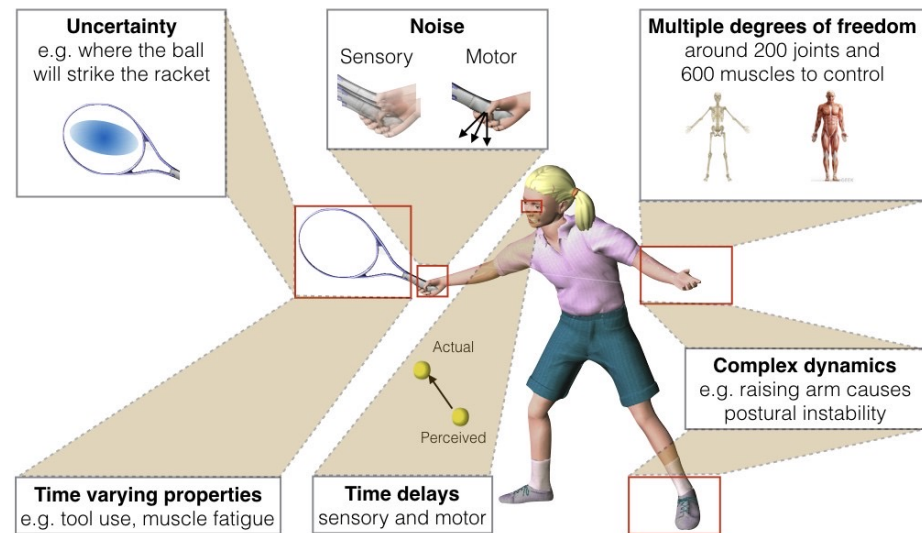
Face-to-phases (2013) - CC-BY Guillaume Dumas

Why do we (and other animals) need a brain?



- To produce adaptable and complex movements: our only way to interact with the world

Module 4 – Motor Control



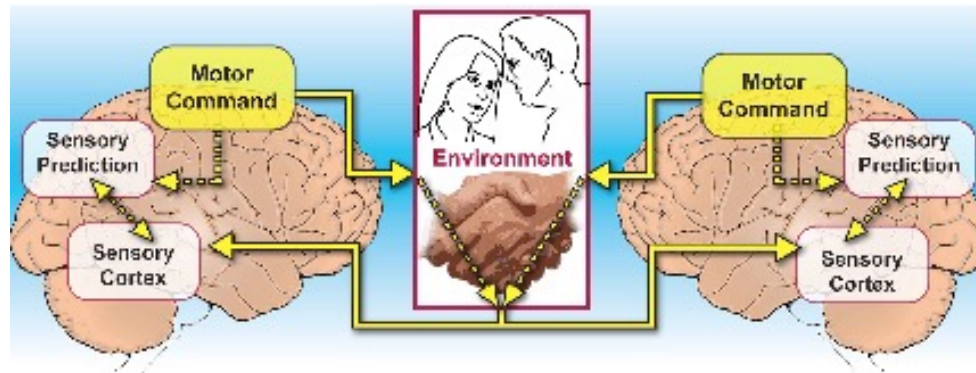
Module 4 – Motor Control

Reading: Lecture notes motor control

Exercises: 3 small sets of class exercises

- You should be able to:
 - explain why motor control is difficult (for humans and AI)
 - describe the concept of internal models, and differentiate between forward and inverse models in motor control
 - explain what forward and inverse models are used for, what their input/output is, and give an example of their use in human motor control
 - apply Bayes rule in action
 - apply decision theory in motor control

Module 5 – Social Cognition



- your action output = another person's perceptual input (and vice-versa)
- what are the cognitive and neural mechanisms that enable us to successfully interact with other people?
- using AI to understand (social) cognitive mechanisms
- project: decoding interactive behaviour/roles from brain (EEG) data

Module 5 – Social Cognition

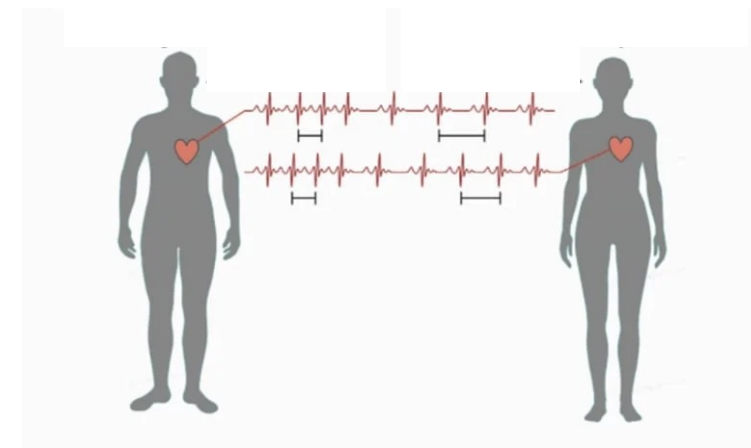
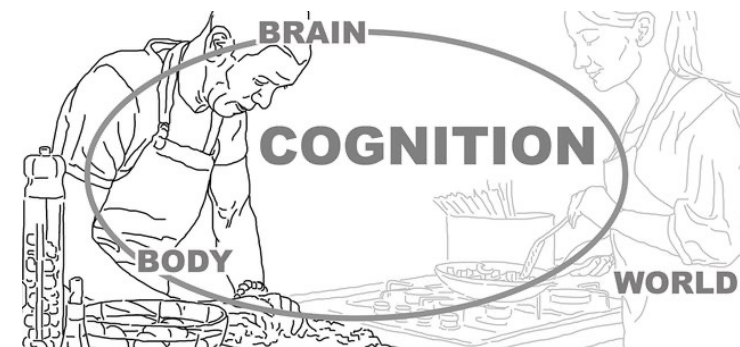
Readings: Lecture notes 8 & 9 (social cognition part I and II)

Exercise: classification of EEG data project

- You should be able to:
 - describe what social cognition is, what makes us social, and why sc is important for AI
 - differentiate between the two main approaches to studying social cognition
 - describe mechanisms of social cognition, e.g., Theory of Mind, and how they are studied/measured in experiments
 - explain the concept and proposed function of the social brain, and be able to discuss this in the context of Bayesian inference
 - describe theories of how we understand others (Theory-Theory & Simulation Theory, mirror neurons)
 - describe mechanisms of joint action
 - describe when two heads are better than one
 - use AI to measure neural mechanisms of social cognition (classification exercise)

Module 6 – Embodied Cognition

- cognition is shaped by aspects of the entire physical body of an agent: it is not restricted to the brain
- it is also shaped by our interactions with the environment and other people
- project:
 - infer social and cognitive processes from bodily signals
 - participate in an interactive experiment, with ECG and movement accelerometry wearables (TODAY!)
 - analyze heart rate variability (HRV) using frequency-domain methods during different interactive games
 - infer friendship from heart rate synchrony



Module 6 – Embodied Cognition

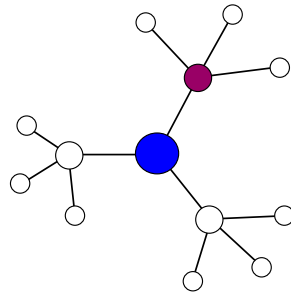
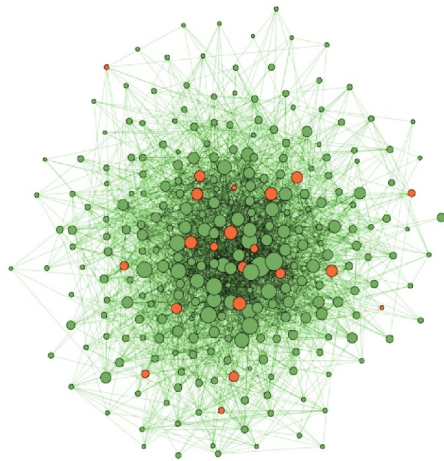
Reading: Wilson, M. (2002). Six views of embodied cognition. *Psychonomic Bulletin & Review* 9, 625-636.

Exercise: class experiment and HRV/heart rate synchrony analysis exercise

- You should be able to:
 - explain the traditional views of cognition, and why these were challenged
 - describe what embodied cognition is, and how this is relevant to AI, including examples from robotics
 - describe the 6 claims of embodied cognition, and be able to give an example of each one
 - discuss problems with some of the claims of embodied cognition
 - describe what psychophysiology is and why embodied cognition underlies this field
 - analyze heart rate variability in the frequency domain, and explain how it relates to human cognition
 - analyze heart rate synchrony, and explain how it informs social cognition

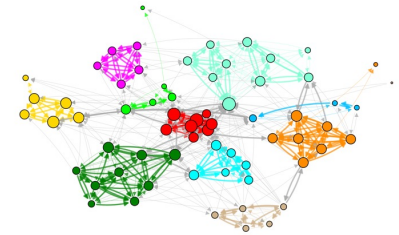
Class & research experiment today!

Motivation: How do the ways in which you connect to each other influence how you interact together in real-time?



Mapping your class social network (in B308)

- We would like to map your class social network

[illegible]

What you have to do ONLINE (please use your laptop, NOT PHONE):

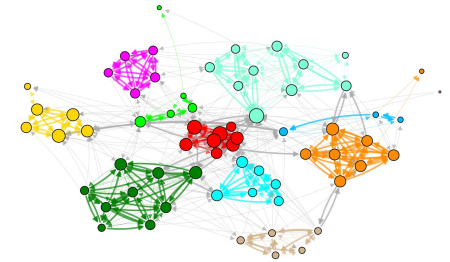
- Read information + Give consent + fill out the social network questionnaire

What you will get:

- Contribution to science!
- You will get to work with your own data later in the course
- Fun + cake
- If you sign up for the subsequent EEG experiment, you will be reimbursed 400 DKK for your participation

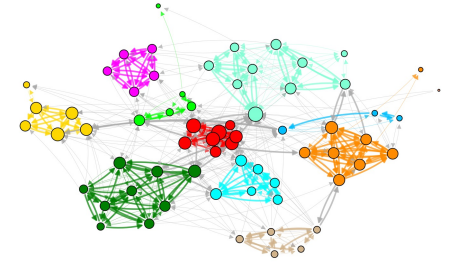
Please use definitions of close friends & friends

- We would like to map your class social network

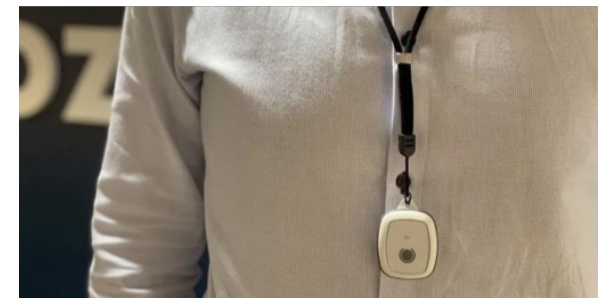
[illegible]

- **“Friends”** are people with whom you like to spend your free time, enjoy informal social activities, such as going for lunch, dinner, drinks, films, etc., and help each other out in a practical way.
- **“Close-friends”** are people with whom you share personal information, provide emotional support, help each other, and enjoy each other’s company.

After the social network questionnaire, we move to B358, rooms 067 & 068

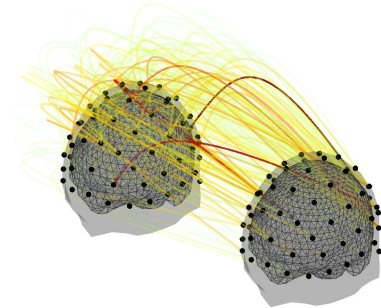


- We will split into 2 groups
- 1 group will do the experiment first, while the other participates in a lecture on “*How does the brain work?*” Then we swap!
- The experiment entails team based icebreaker games!
- But we will also record your heart rates, physical proximity, and movement accelerometry during the games



Research study: EEG experiments

- We will invite you to participate in an EEG interaction experiment
- You will receive an invitation shortly after you have declared your interest



You will be reimbursed for participation (400 dkk)